

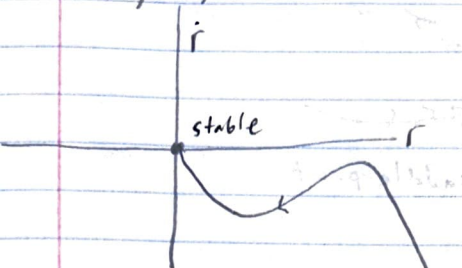
8.4 Cycle Bifurcations

= Global!

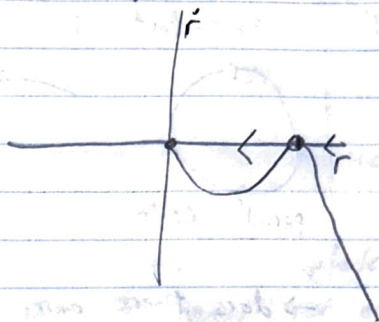
system $\begin{cases} \dot{r} = \mu r + r^3 - r^5 \\ \dot{\theta} = \omega + br^2 \end{cases}$

radial, uncoupled

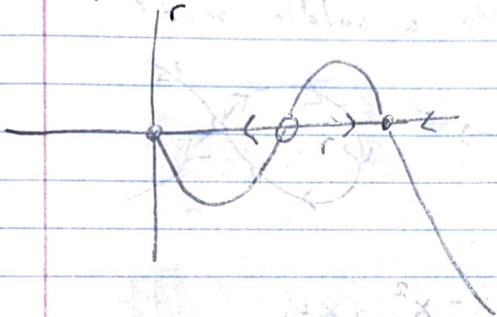
$\mu < \mu_c$



$\mu = \mu_c$

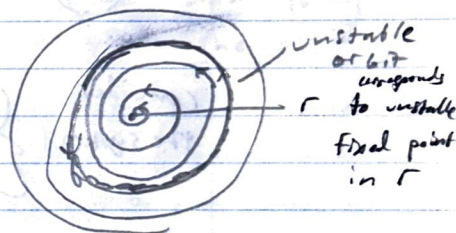


$\mu > \mu_c$



$\mu_c = \frac{1}{2}$

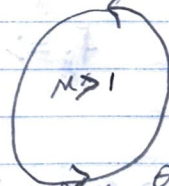
- Similar to a blue-sky bifurcation
- attractive FP always @ the origin
- In polar coordinates:



Josephson Junction

$\begin{cases} \dot{r} = r(1-r^2) \\ \dot{\theta} = \mu - \sin \theta \end{cases}$
A/C external E field

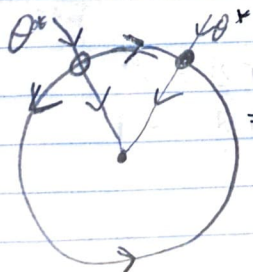
$\theta \approx 0$
Slow $\theta \approx 90^\circ (\frac{\pi}{2})$



Fast $\theta \approx 270^\circ (\frac{3\pi}{2})$

$\theta \approx \pi$

$\mu_c = 1$ since $\sin \theta \in [-1, 1]$



$\mu < \mu_c$

$\Rightarrow \theta^* = \pm \sin^{-1} \mu$

radial eqn causes inflow along θ^* lines

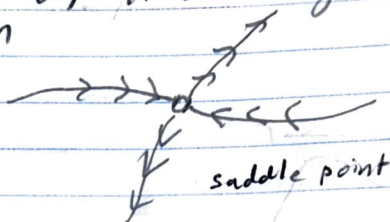
infinite period bifurcation

Homoclinic Bifurcation

→ Suppose you have a bifurcation w/ saddle points, fixed points w/ linear eigenvectors of opposite orientation

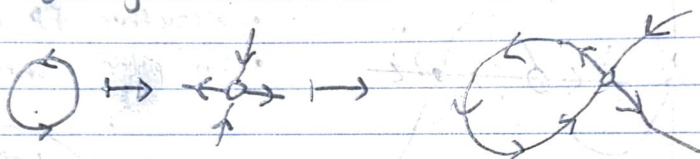


periodic orbit



saddle point

→ does there exist a bifurcation where a periodic orbit topologically deforms into a saddle point?



consider the system

$$\begin{cases} \dot{x} = y \\ \dot{y} = \mu y + x - x^2 + xy \end{cases}$$

→ Find equilibrium (x^*, y^*) :

$$f_x \Rightarrow y^* = 0 \Rightarrow x^* \rightarrow x^* - x^{*2} = 0$$

$$\Rightarrow x^* = (0, 1)$$

$$y^* = (0)$$

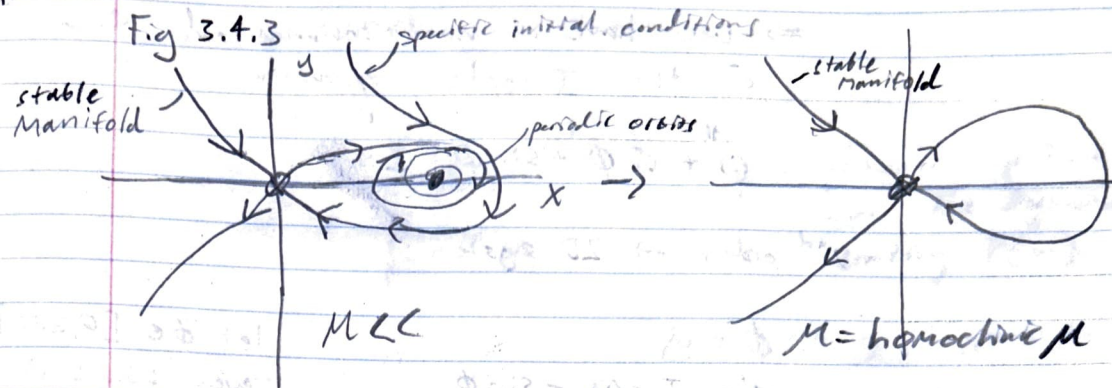
→ construct matrix of velocity gradients (neglecting $O(x^*, y^*)$)

$$A = \begin{pmatrix} 0 & 1 \\ 1 & \mu \end{pmatrix}$$

$$\Rightarrow \text{Tr } A = \mu$$

$$\det A = -1$$

Fig 3.4.3



→ takes infinite time to reach the complete saddle orbit, since derivatives go to zero at origin

→ however, it is possible to recast the time dimension to depend on the position, so that the "infinite time" is circumvented and you can compute how long that trajectory will take to complete the orbit

8.5 Driven Pendulum

capacitance

friction term

constant current

$$\frac{\hbar C}{2L} \ddot{\phi} + \frac{\hbar}{2eR} \dot{\phi} + I_c \sin \phi = I_B$$

resistance

importance of opposing the E field.

$$\text{let } I = \frac{I_B}{I_c}$$

$$\Rightarrow \frac{\hbar C}{2eI_c} \ddot{\phi} + \frac{\hbar}{2eRI_c} \dot{\phi} + \sin \phi = I$$

$$\text{let } \tau \rightarrow \left(\frac{2eI_c}{\hbar C} \right)^{1/2} t$$

$$\Rightarrow \ddot{\phi} + \underbrace{\left(\frac{\hbar}{2eRI_c} \right) \left(\frac{\hbar C}{2eI_c} \right)^{1/2}}_{\equiv \alpha} \dot{\phi} + \sin \phi = I$$

$\Rightarrow$ we have a nondimensionalized form of the Josephson equation

$$\phi'' + \alpha \phi' + \sin \phi = I$$

2nd order $\rightarrow$ 2D system

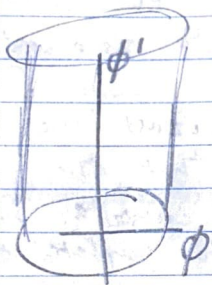
$$\phi' = y$$

$$y' = I - \alpha y - \sin \phi$$

let $\phi \in [0, 2\pi]$

even though ϕ is a compact variable,

ϕ' can take any value



$$(\alpha \geq 0, I > 0)$$

$\rightarrow$ in the case $\alpha = 0$, the system becomes a pendulum that conserves energy

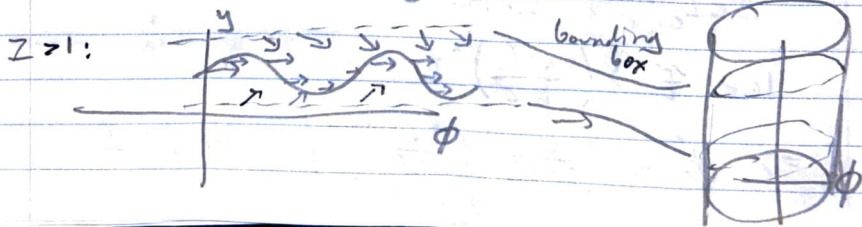
$\rightarrow$ Equilibria for $\alpha \neq 0$:

$$y^* = 0, \rightarrow \phi^* = \sin^{-1}(I)$$

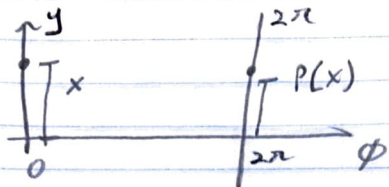
$I > 1$: no critical points

$I \leq 1$: 2 critical points

$\rightarrow$ Matrix of stability $\Rightarrow$ either a saddle or a sink



Poincaré Map / First return map



$P(x)$ is a Poincaré map that tells you how the dynamics work by iterating $P(y)$

$P(x)$

